MATH 130 · UNIT 3B

Trigonometric Functions of Any Angle

Concepts, worked examples, student practice, and synthesis

55 minutes · Offline capable

Today’s Learning Roadmap

1. find reference angles in degrees and radians
2. determine trigonometric signs and quadrants
3. find exact trigonometric values from points and reference angles, and solve sine and cosine equations
4. apply the trigonometric triangle-area formula

Predict: find reference angles in degrees and radians

PREDICT FIRST

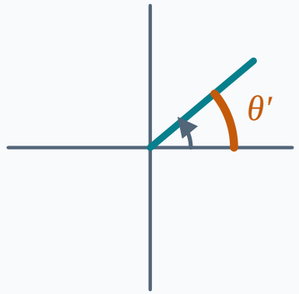
Why is a reference angle always acute?

Commit to a response before discussion.

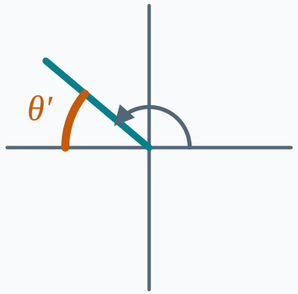
find reference angles in degrees and radians

FORMULA FOCUS

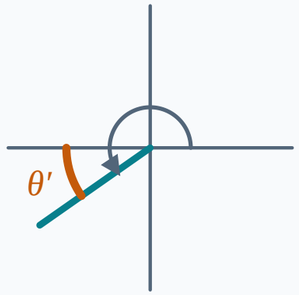
- Reference angles are measured to the nearest x-axis.
- First find a coterminal angle when the original lies outside one rotation.



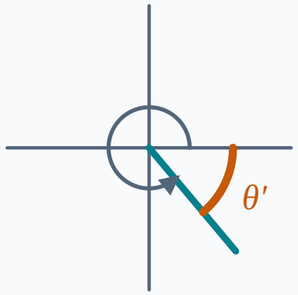
Quadrant I: θ′ = θ



Quadrant II: θ′ = 180° − θ



Quadrant III: θ′ = θ − 180°



Quadrant IV: θ′ = 360° − θ

Measure the reference angle to the nearest part of the x-axis.

Model the Skill: find reference angles in degrees and radians

1. 150° has reference angle 30°; 310° has 50°.

2. -120° is coterminal with 240°, so its reference angle is 60°.

3. $7\pi/4$ has reference angle $\pi/4$.

You Try: find reference angles in degrees and radians

STUDENT PRACTICE

Find the reference angle for 580°.

Reveal solution

FEEDBACK AND SOLUTION

$580 - 360 = 220^\circ$, so the QIII reference angle is 40°.

Check and Diagnose: find reference angles in degrees and radians

ERROR ANALYSIS

- Do not measure to the y-axis.
- Use π, not 180°, when working in radians.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: find reference angles in degrees and radians

PACING CHECKPOINT

State the quadrant formulas for reference angles.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Predict: determine trigonometric signs and quadrants

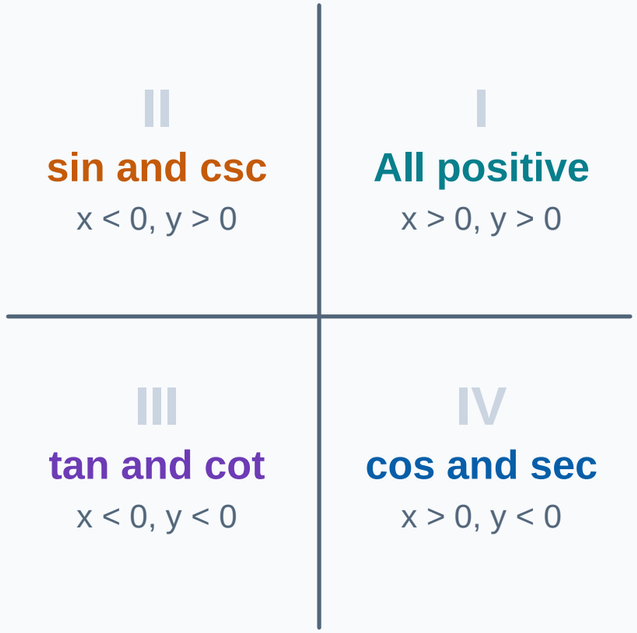
PREDICT FIRST

If tangent is positive and cosine is negative, which quadrant is possible?

Commit to a response before discussion.

determine trigonometric signs and quadrants

1. ASTC summarizes positive functions by quadrant.
2. Signs follow from x, y, and positive r.



sin = y/r, cos = x/r, tan = y/x, and r is always positive

The signs of x and y decide each function’s sign.

Source: 10

Model the Skill: determine trigonometric signs and quadrants

Determine the quadrant if $\tan\theta > 0$ and $\cos\theta < 0$.

1. Tangent is positive in Quadrants I and III.
2. Cosine is negative in Quadrants II and III.
3. The intersection is Quadrant III.

Source: Authored

You Try: determine trigonometric signs and quadrants

STUDENT PRACTICE

Determine the quadrant if $\sin\theta > 0$ and $\sec\theta < 0$.

Reveal solution

FEEDBACK AND SOLUTION

Quadrant II.

Check and Diagnose: determine trigonometric signs and quadrants

Use the intersection of sign conditions.

Reciprocal functions share the sign of their partner.

Source: Authored

Make It Stick: determine trigonometric signs and quadrants

PACING CHECKPOINT

Explain ASTC using the signs of x and y rather than memorization.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Predict: find exact trigonometric values from points and reference angles

PREDICT FIRST

What does r represent for a point (x, y) on a terminal side?

Commit to a response before discussion.

find exact trigonometric values from points and reference angles

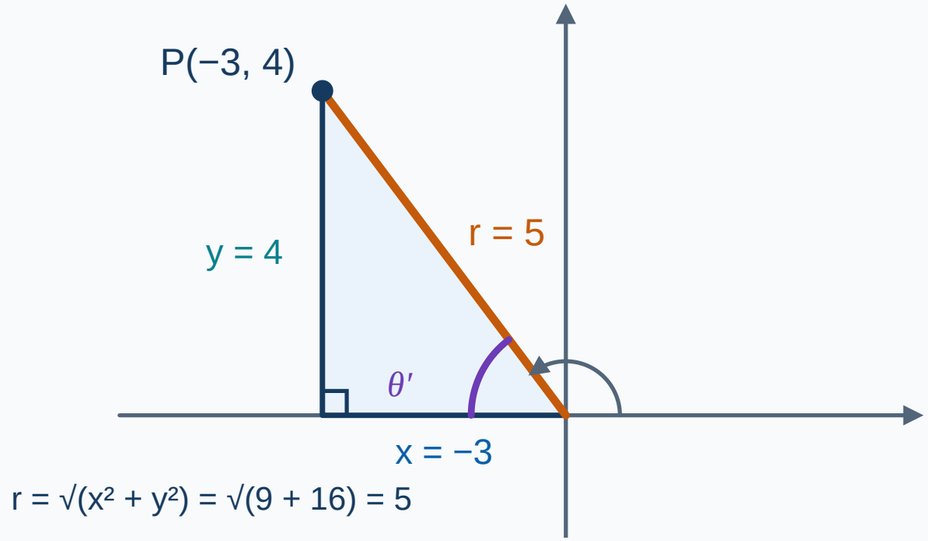
CONCEPT

- $r = \sqrt{x^2 + y^2}$.
- Use the reference angle for magnitude and the quadrant for sign.

Model the Skill: find exact trigonometric values from points and reference angles

For $P = (-3, 4)$, calculate r and all six trigonometric values.

1. $r = \sqrt{(-3)^2 + 4^2} = 5$.
2. $\sin\theta = 4/5$, $\cos\theta = -3/5$, $\tan\theta = -4/3$.
3. $\csc\theta = 5/4$, $\sec\theta = -5/3$, $\cot\theta = -3/4$.



r comes from the Pythagorean theorem and is always positive.

You Try: find exact trigonometric values from points and reference angles

STUDENT PRACTICE

Find exact trig values for $P = (-2, -5)$ and evaluate $\sin 240^\circ$.

Reveal solution

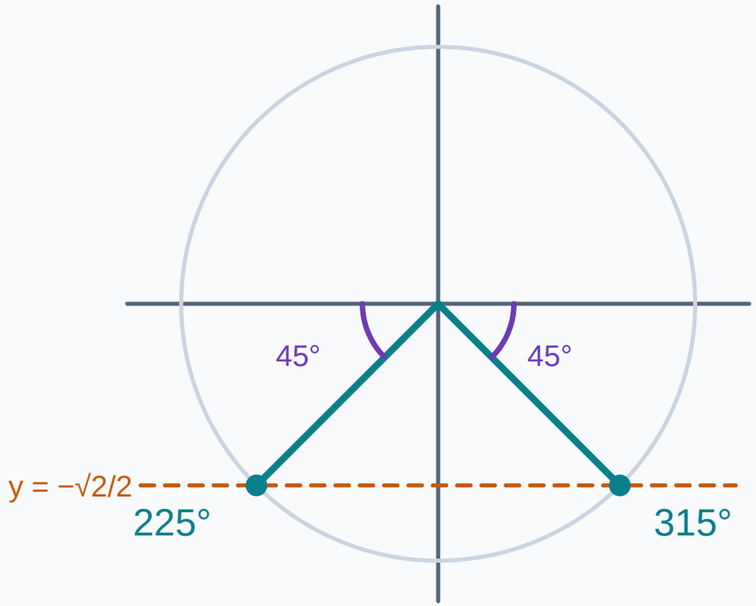
FEEDBACK AND SOLUTION

- For $P = (-2, -5)$, $r = \sqrt{29}$; $\sin = -5/\sqrt{29}$, $\cos = -2/\sqrt{29}$, $\tan = 5/2$.
- $\csc = -\sqrt{29}/5$, $\sec = -\sqrt{29}/2$, $\cot = 2/5$; $\sin 240^\circ = -\sqrt{3}/2$.

Model the Skill: solve a sine or cosine equation

Find all θ in $[0^\circ, 360^\circ)$ with $\sin\theta = -\frac{\sqrt{2}}{2}$.

1. Reference angle: $\sin\theta' = \frac{\sqrt{2}}{2}$ gives $\theta' = 45^\circ$.
2. Sine is negative in Quadrants III and IV.
3. $\theta = 180^\circ + 45^\circ = 225^\circ$ and $\theta = 360^\circ - 45^\circ = 315^\circ$.



Two terminal sides meet the line y = −√2/2.

Source: Authored

You Try: solve a cosine equation

STUDENT PRACTICE

Find all θ in $[0^\circ, 360^\circ)$ with $\cos\theta = -\frac{\sqrt{2}}{2}$.

Reveal solution

FEEDBACK AND SOLUTION

Reference angle 45°; cosine is negative in Quadrants II and III, so $\theta = 135^\circ$ and $\theta = 225^\circ$.

Source: Authored

Check and Diagnose: find exact trigonometric values from points and reference angles

ERROR ANALYSIS

- r is always positive.
- Rationalize denominators only when required by course convention.
- An equation like $\sin\theta = k$ has two answers on one rotation; find both.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: find exact trigonometric values from points and reference angles

PACING CHECKPOINT

Describe the point-to-ratios workflow in three steps.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Predict: apply the trigonometric triangle-area formula

PREDICT FIRST

Why must the known angle be included between the two known sides?

Commit to a response before discussion.

apply the trigonometric triangle-area formula

CONCEPT

- $A = \frac{1}{2}bc\sin A$ uses two sides and their included angle.
- The formula applies to nonright triangles.

Model the Skill: apply the trigonometric triangle-area formula

Find the area when $A = 76^\circ$, $b = 34$, and $c = 21$.

1. $A_{triangle} = \frac{1}{2}bc\sin A$.
2. $A_{triangle} = \frac{1}{2}(34)(21)\sin 76^\circ \approx 346.40$.
3. The result is in square units.

You Try: apply the trigonometric triangle-area formula

OPTIONAL

STUDENT PRACTICE

Find the area when sides 12 and 15 include a 40° angle.

Reveal solution

FEEDBACK AND SOLUTION

$A = \frac{1}{2}(12)(15)\sin 40^\circ \approx 57.85.$

Source: Authored

Check and Diagnose: apply the trigonometric triangle-area formula

OPTIONAL

ERROR ANALYSIS

- Match each side pair with its included angle.
- Square units are required.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: apply the trigonometric triangle-area formula

PACING CHECKPOINT

Explain how the diagram identifies the correct sine angle.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Source: Authored

Session Summary and Exit Ticket

1. State the most important idea from today without looking at your notes.
2. Complete one representative setup and identify the step most likely to cause an error.
3. Write one question that should be answered before the next assessment.

Exit response

Answer in complete mathematical sentences and identify one remaining question.

Reveal review guidance

FEEDBACK AND SOLUTION

Use the objective roadmap to identify any skill that still needs deliberate practice.